

muchísimocks: An emulated cosmological simulation suite for simulation-based inference

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ABSTRACT

Galaxy clustering contains rich cosmological information down to non-linear scales; however, methods to extract this information are currently limited by the high cost of cosmological N -body simulations, as well as by the fidelity of galaxy bias modeling. We present a new approach to generating approximate but robust galaxy mocks: we employ the `map2map` full-field emulator to generate high-fidelity displacement fields, and apply hybrid Lagrangian bias expansion approach to flexibly model biased tracers. With this approach, we construct a suite of mock 3D galaxy density fields each of size $(1\ h^{-1}\ \text{Gpc})^3$ at $z = 0$ spanning 10,000 different cosmologies, called **muchísimocks**. To demonstrate the power of this data set, we use it to perform simulation-based inference (SBI) of galaxy clustering. We train an SBI model on summary statistics of the density fields—the galaxy–galaxy power spectrum, the galaxy–matter power spectrum, and the galaxy bispectrum—and infer the cosmological parameters, marginalizing over the bias and noise coefficients. We validate the pipeline on in-distribution test mocks and find our inferred posteriors to be well-calibrated across the prior range. With this large suite, we are able to test convergence with the number of simulations; we find that while 20,000 training points are sufficient for the power spectrum, when including the additional statistics precision continues to improve beyond 320,000, and that increasing the number of bias parameter sets per cosmology yields similar constraints as increasing the number of cosmologies. We also show that we recover unbiased cosmological constraints from an out-of-distribution model, an independent high-resolution N -body simulation populated with galaxies using extended SubHalo Abundance Matching, demonstrating the generalizability of the approach. We find that the galaxy–matter power spectrum is critical for breaking parameter degeneracies, and additionally including the bispectrum significantly improves precision, on σ_8 by XXX and b_1 by XXX. This paves the way for exploring full-field inference from galaxy clustering, as well as a range of other potential applications. The **muchísimocks** suite and the code used in this paper are available at [thisurl](#). [ksf: emphasize to help guide design of future sbi approaches; in abstract, and section of paper](#) [ksf: maybe include 3x2pt / pgm in abstract](#)

Key words: methods: numerical – methods: statistical – galaxies: statistics – cosmology: large-scale structure of Universe – cosmological parameters

1 INTRODUCTION

The clustering of galaxies encodes information about the cosmological model describing the universe. Surveys such as the Sloan Digital Sky Survey ? and most recently the Dark Energy Spectroscopic Instrument ? have spectroscopically mapped millions of galaxies, producing detailed maps of a large volume of the observable universe. Upcoming surveys such as the Roman Observatory ? will continue delivering observations . . .

Standard approaches to analyzing this spectroscopic survey data rely on analytic techniques to modeling summary statistics of the galaxy field ?. Analyses largely focus on two-point statistics, which contain robust cosmological information but do not fully capture the information content in the data.

Simulation-based inference (SBI) has shown promise for cosmological inference from galaxy clustering. In the SBI approach, one forward models the data based on a prior distribution of the parameter space, and uses a distance metric to select samples that match the data. This has several advantages compared to standard Monte Carlo-based inference approaches. First, analytically modeling statistics beyond the power spectrum is difficult; with SBI, one can forward model the full galaxy field and directly measure any statistic. Similarly, modeling the effect of systematics on the statistics is difficult, particularly as more subtle systematic effects become important with improving data; with SBI, these systematics can be forward-modeled and the statistics simply measured. Finally, Monte Carlo methods typically assume a Gaussian form for the likelihood (and must estimate a nontrivial covariance matrix), even though particularly higher-order statistics are known to have non-Gaussian properties; SBI makes no assumptions about the form of the likelihood.

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While promising, SBI faces challenges for galaxy clustering analyses. One key issue is that SBI typically requires many simulations, but cosmological simulations are very computationally expensive. An N -body simulation with the volume and resolution to accurately describe galaxy clustering into the nonlinear regime costs XXX CPU hours, and thousands to hundreds of thousands of simulations are generally needed for SBI. Another challenge is modeling how galaxies populate the dark matter distribution; models must be relatively fast, flexible enough to model various galaxy populations, and accurate down to small scales.

The SimBIG collaboration has demonstrated the SBI approach for galaxy clustering, achieving constraining power beyond standard analyses. They use the 2,000 simulations of the Quijote suite, and a halo occupation distribution (HOD) model to populate the simulations with galaxies. [others]

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In this paper...

2 THE MUCHÍSIMOCKS SUITE

2.1 Emulating the displacement field with map2map

The nonlinear matter distribution can be simulated to high accuracy with N -body simulations. However, simulations with the resolution and volume required to analyze current galaxy surveys are very expensive, and training simulation-based inference models requires a very large of simulations. We thus turn to a field-level emulator of the matter distribution, map2map.

The density field can be approximated at linear order by the Zel’dovich approximation (ZA, also known as first-order Lagrangian perturbation theory, ILPT), based on a realization of an initial conditions field. The map2map emulator learns a mapping from the ZA field at $z = 0$ to the non-linear N -body displacement field at that redshift, effectively learning the correction from linear to non-linear displacements. The emulator is trained on 2000 simulations of the Quijote simulation suite that span a latin hypercube (LH) of five cosmological parameters: Ω_c , σ_8 , h , Ω_b , and n_s . Each has a volume of $(1 h^{-1} \text{Gpc})^3$ with 512^3 particles, giving particle mass resolutions in the range $2 \times 10^{11} - 1 \times 10^{12} h^{-1} M_\odot$. map2map also trains a separate network to predict the nonlinear velocity field from the ZA velocities. Once trained, map2map can generate the displacement and velocity fields for any input ZA field, taking a few minutes on a GPU for a $(1 h^{-1} \text{Gpc})^3$ volume. While we do not utilize the velocity fields in this work, as we limit to real-space analysis, these will be important for future analyses including redshift-space distortions as well as other applications involving the velocity distribution; we release them as part of the muchísimocks suite. [ksf: explain special velocity field weights used, for zspace](#)

In more detail, map2map consists of two convolutional neural networks (CNNs), one to predict the matter density field and the other to predict the velocity field. The particle displacements are structured on a 3D grid as they are functions of their initial Lagrangian space position, so the CNN processes the inputs and outputs as 3-channel images. As most of the cosmological parameters only affect the input power spectrum that is used to create the ZA field, these are not needed as additional inputs to the model. However, the matter density Ω_m affects the N -body evolution, so is included as an input “style” parameter; this is incorporated via the approach of the StyleGAN2, in which the style parameters modulate the convolution kernels. The model uses a V-net-like architecture, with 3 layers of down-sampling and up-sampling, and leaky ReLU activations. For its loss

function, it combines the log mean squared error of the Lagrangian particle displacements and Eulerian density field.

For our mock suite, we define a latin hypercube of 10,000 sets of cosmological parameters: Ω_c , σ_8 , h , Ω_b , and n_s . We use uniform priors for each parameter that span ~ 10 around the values of the cosmological parameters, following (with a range for σ_8 extended to slightly lower values); these are shown in Table 1. For each of these, we generate a ZA field based on a different random seed, and pass that field through map2map to obtain the nonlinear particle displacements. The map2map-emulated particle displacements are a key ingredient in modeling the galaxy bias fields, as discussed in the next section.

2.2 Modeling tracers with the hybrid bias expansion

To model biased tracers of the matter field, we use the hybrid Lagrangian bias expansion, also known as hybrid effective field theory (HEFT, ???). We first define a mapping from initial (Lagrangian) space coordinates $\mathbf{q}$ to Eulerian space coordinates $\mathbf{x}$ as $\mathbf{x} = \mathbf{q} + \Psi(\mathbf{q})$, where $\Psi(\mathbf{q})$ is the Lagrangian space displacement field. This advection from Lagrangian to Eulerian space effectively expresses the overdensity field as weighted contributions from the initial density field that end up in a given Eulerian location. The displacement term $\Psi(\mathbf{q})$ can be computed with Lagrangian perturbation theory, but this is only accurate down to mildly non-linear scales; here we take the “hybrid” approach which uses N -body displacements to compute $\Psi(\mathbf{q})$, significantly improving the accuracy down to non-linear scales.

The galaxy overdensity field in Lagrangian space can be described to second order as

$$\delta_g(\mathbf{q}) = 1 + b_1 \delta_L(\mathbf{q}) + b_2 \left[\delta_L^2(\mathbf{q}) - \langle \delta_L^2 \rangle \right] + b_{s^2} \left[s^2(\mathbf{q}) - \langle s^2 \rangle \right] + b_\Delta \nabla^2 \delta_L(\mathbf{q}) \quad (1)$$

where $\delta_L(\mathbf{q})$ is the linear matter density in Lagrangian space, and $s^2(\mathbf{q})$ is the shear field, defined by the traceless part of the tidal field, given by $s^2(\mathbf{q}) = s_{ij}(\mathbf{q}) s^{ij}(\mathbf{q}) = \delta_i \delta_j \phi(\mathbf{q}) - \frac{1}{3} \delta_{ij}^K \delta_L^2(\mathbf{q})$, where $\phi(\mathbf{q})$ is the linear gravitational potential in Lagrangian space $\nabla^2 \phi(\mathbf{q}) = 4\pi G \bar{\rho} \delta_L(\mathbf{q})$, and δ^K is the Kronecker delta. The coefficients b_1 , b_2 , b_{s^2} , and b_Δ are the Lagrangian bias parameters. The prior ranges of the bias parameters used in our SBI analysis are shown in Table 1, though we note that a tracer field could be constructed for any choice of bias parameters with our saved bias fields.

We build this field starting from a random realization of the initial Gaussian field on a $N_{\text{grid}} = 512^3$ grid in a $L = (1 h^{-1} \text{Gpc})^3$ box, matching the resolution and size of the Quijote simulations that map2map was trained on, and use the ZA to compute approximate displacements. We pass the ZA displacement and velocity fields as input to map2map (§2.1), which outputs emulated N -body displacement and velocity fields. We compute the Lagrangian operators based on the linear density field, smoothing these with a damping scale of $k = 0.75 h \text{Mpc}^{-1}$. We then perform the advection to Eulerian space: The tracers are deposited onto a grid in Eulerian space with a cloud-in-cell (CIC) assignment, with positions given by the simulation displacement field and weights given by Lagrangian operator. This gives five bias fields in Eulerian space, O_i : weighting the Lagrangian density field by the constant 1 gives the homogeneous field; by $\delta_L(\mathbf{q})$, the linear density field; by $\delta_L^2(\mathbf{q}) - \langle \delta_L^2 \rangle$, the quadratic density field; by $s^2(\mathbf{q}) - \langle s^2 \rangle$, the tidal field; and by $\Delta^2 \delta(\mathbf{q})$, the Laplacian. These fields can be weighted by the bias coefficients and summed to produce an overall tracer field δ_g .

We perform two postprocessing steps on these fields. [ksf: fill in To](#)

Figure 1 placeholder: schematic (pipeline overview)

Figure 1. Schematic overview of the muchisimocks pipeline. [ksf: placeholder](#)

reduce storage space, we reduce the size of these fields by cutting the high- k modes: we remove modes beyond $k_{\text{Nyquist}} = \pi N_{\text{grid, cut}}/L \approx 0.4$, where $N_{\text{grid, cut}} = 128$, our reduced grid size. We do this by Fourier transforming each bias field, nulling the modes higher than k_{Nyquist} , and inverse Fourier transforming back to real space. At k scales higher than this our hybrid bias model loses accuracy anyway and we don't consider them in our inference, so we do not give up accuracy with this k -cut. We perform the same reduction on the velocity fields, cutting each component of the velocity vectors in k -space. We finally perform a deconvolution of the bias fields, using a linear interpolation kernel at the original grid size N_{grid} to correct for the smoothing by the CIC kernel. (Technically the deconvolution should be performed before the k -cut, but the results are effectively equivalent and this ordering is much faster.) This deconvolution could be done at the summary statistics level, but we choose to deconvolve the fields for ease of use.

The final main suite, designed as a training set, contains 10,000 sets of tracer position fields (5 Lagrangian bias fields advected to Eulerian space) and velocity fields, each at a different cosmology; each is of size $(1h^{-1} \text{ Gpc})^3$ and resolution 128^3 . We also generate multiple smaller suites for testing, detailed in §3.5. We call these suites *muchisimocks* (a portmanteau of the words “muchísimos”, Spanish for “many”, and “mocks”).

2.3 Modeling noise at the field level

[ksf: technically this is a postprocessing step only needed for the SBI part, where should it go?](#)

We require a model for stochastic noise in order to model realistic discrete galaxy populations. This can be modeled at the level of each summary statistic, but our goal is to make this suite usable for a wide range of statistics as well as field-level analyses; we thus desire a field-level noise model. Further, as we are considering higher-order statistics, we need to include non-Gaussian noise contributions. We thus turn to the noise model of ?, an EFT approach described by nonlinear couplings of a single Gaussian field to the bias fields. In

this model the galaxy density field can be written as

$$\delta_g(\mathbf{x}) = \sum_{m=0}^{\infty} \sum_{\mathcal{O}} b_{\mathcal{O}}^{\{m\}} [\epsilon_G(\mathbf{x})]^m \mathcal{O}(\mathbf{x}) \quad (2)$$

where m is the order of the expansion in the noise field, $\epsilon_G(\mathbf{x}) = \mathcal{N}(0, 1)$ is a unit Gaussian field, $\mathcal{O}$ are operators of the galaxy density field, and $\{b_{\mathcal{O}}^{\{m\}}, b_{\mathcal{O}}^{\{m\}}\}$ are free coefficients. In our case we limit to second order in the bias expansion, as described in §2.2; limiting to zeroth order in m reduces to the noiseless case, as in Equation 1. For our SBI analysis, we include terms through first order in the noise expansion, $m = 0, 1$. That gives us bias coefficients $\{b_{\mathcal{O}}^{\{0\}}\} = b_1, b_2, b_{s^2}, b_{\Delta}$, as in Equation 1 (we fix $b_{\mathcal{O}}^{\{0\}} = 1$), and noise coefficients $\{b_{\mathcal{O}}^{\{1\}}\} = A_{\text{homog}}^{\text{noise}}, A_{b_1}^{\text{noise}}, A_{b_2}^{\text{noise}}, A_{b_{s^2}}^{\text{noise}}, A_{b_{\Delta}}^{\text{noise}}$.

We note that limiting the noise model to $m = 1$ means that we exclude noise contributions to $P_{gm}(k)$, as at $m = 1$ the noise field ϵ_G has zero mean (whereas $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ get factors of ϵ_G^2 which don't vanish). This means that the bias parameters may have to absorb partial noise contributions, though we show that we still achieve largely unbiased bias parameters; future work could extend this noise model to $m = 2$ or choose a well-motivated subset of noise parameters tailored to the statistics analyzed.

Operationally, we generate one unit Gaussian noise field ϵ_G per cosmology in our suite. We store these and use them to construct the noisy tracer field on-the-fly just before computing the summary statistics of the field. This means that we use the same noise realization when we select different bias and noise parameter sets for the same cosmology (detailed in §3.2 and §3.5), which introduces some undesirable non-independence; however, we expect the effects of this to be minimal, and the storage cost of one noise field per bias model is prohibitive. The prior ranges of the noise parameters used in our SBI analysis are shown in Table 1.

3 SIMULATION-BASED INFERENCE METHODOLOGY

3.1 Neural posterior estimation

The main idea behind simulation-based inference is that we can infer a posterior distribution of our data over the parameters of interest using only a *simulator*, which produces data samples given a set of parameters. This is distinct from explicit likelihood inference, in which we are required to define a likelihood of our data with respect to the parameters. Neural density estimators (NDEs) present a flexible and robust way to model a target distribution. Here, we use neural posterior estimation, which uses an NDE to directly model the posterior distribution.

To do this, we aim to minimize the distance between the true posterior $p(\theta|x)$ and the NDE model of the posterior $q_\phi(\theta|x)$, where ϕ are the parameters of the neural network model. We use maximum likelihood estimation, which minimizes the Kullback-Leibler divergence between the distributions. In practice, we consider the posteriors multiplied by $p(x)$:

ksf: update to using align

$$E = \dots \quad (3)$$

$$= L \quad (4)$$

where in the second line, we have converted the integral to a sum as we are using discrete samples from our simulation; in the third line, we observe that we are minimizing with respect to the neural network parameters ϕ and the true posterior $p(\theta|x)$ is independent of ϕ so it drops out; and in the third line we use the fact that minimizing the positive is the same as maximizing the negative.

Normalizing flows are a popular choice for the NDE, thanks to their expressivity and flexibility. The key idea is to transform an initial simple distribution into the target distribution through a chain of transformations. Specifically we use masked autoregressive flows (MAFs), which stack Masked Autoencoders for Distribution Estimation (MADEs) in order to model more complex distributions. This is defined as

$$q_\phi(\theta|x) = \dots \quad (5)$$

The MAF samples an initial base variable z_0 from a Gaussian N , and then performs the transformations $z_t = f_t(z_{t-1}, \theta)$, where f_t is neural network that takes as input the previous variable z_t as well as the parameters θ . The Jacobian (the determinant term) arises from the change of variables, and involves f^{-1} , thus requiring f to be invertible. The invertibility property of NFs both enables us to sample from the network and gives us an explicit density, which is key for performing SBI. ksf: via [tucci paper](#), check The loss function derived from the KL divergence (Equation 4) is used to train the weights and biases of the network.

3.2 Training set

To train our simulation-based inference model, we use the main *muchisimocks* suite of bias fields to generate a set of tracer fields. Our prior on the cosmological parameters is set by the range of the suite, as described in 2.1. We need to define the range of the bias coefficients $\{b\}$ we use to sum the fields; for this, we use the results of ?, which define priors that capture a range of realistic galaxy populations modeled with the extended subhalo abundance matching technique ?. We choose to use uniform priors with slightly wider ranges than the relations found by ?, a more conservative choice; these prior ranges are shown in Table 1. For the noise coefficients

Table 1. Uniform prior ranges for the cosmological, bias, and noise parameters. The cosmological parameters are based on a ~ 10 range around the ? best-fit values, and the bias and noise parameters are loosely based on the hypervolume found by ? for a wide range of realistic galaxy populations.

Parameter	Lower bound	Upper bound
<i>Cosmological parameters</i>		
Ω_c	0.23	0.40
σ_8	0.65	0.90
h	0.6	0.8
Ω_b	0.04	0.06
n_s	0.92	1.01
<i>Bias parameters</i>		
b_1	-1.0	3.0
b_2	-2.0	2.0
b_{s^2}	-2.0	2.0
b_Δ	-10.0	10.0
<i>Noise parameters</i>		
$A_{\text{homog}}^{\text{noise}}$	-3.0	3.0
$A_{b_1}^{\text{noise}}$	-3.0	3.0
$A_{b_2}^{\text{noise}}$	-2.0	2.0
$A_{b_{s^2}}^{\text{noise}}$	-5.0	5.0
$A_{b_\Delta}^{\text{noise}}$	-10.0	10.0

$\{A^{\text{noise}}\}$ we choose wide uniform priors loosely based off of the corresponding (noiseless) bias coefficients.

While we are limited to the 10,000 cosmologies defined in the *muchisimocks* suite, we can combine these into tracer fields with any number of bias parameter sets we choose. At a given cosmology, we then have multiple tracer fields generated with distinct sets of bias and noise parameters. Though this means the final tracer fields are not fully independent of each other, this has been shown (e.g. in ?) to improve inference by significantly augmenting the training data size and allowing the model to disentangle cosmology and bias parameters. Note that the summing of bias fields with the chosen bias coefficients is essentially instantaneous and can be done in postprocessing just before computing summary statistics, and thus requires no additional compute time or storage.

As we later want to test convergence of the inference with the number of bias parameter sets, we use a nested latin hypercube to generate the bias parameters. The top-level LH contains 320,000 points, resulting in 32 independent bias parameter sets for each of the 10,000 cosmologies. Nested subsets of this LH are also LHs, with 16, 8, 4, 2, and 1 bias parameter sets per cosmology. This was achieved with the ? algorithm for nested latin hypercubes, which works by ... ksf: TODO. A python implementation of this algorithm along with validation tests are available in the *gethypercube*¹ package. ksf: should i include an appendix detailing the nested-LH package? or just link to the github as im doing? or does it deserve it's own little software paper? This is shown to achieve LHs at each level that have properties much closer to a true LH than either taking random subsets of a top-level LH, or stacking bottom-level LHs. The use of LH-distributed parameters has been shown to be important for accurate and robust training, so we expect that this nested strategy leads to robustness in our convergence tests, which are a key result of this work. We generate one nested LH for the noiseless case with the 4-parameter set of bias parameters; and a separate nested LH for

¹ <http://github.com/kstoreyf/gethypercube>

the noisy case with 4 bias parameters and 5 noise parameters, for a total of 9 independently varied parameters.

To generate our final tracer fields, we multiply the muchisimocks bias fields and noise field by the selected set of corresponding bias and noise parameters as in Equation (2), and sum these weighted fields. We normalize these fields by the original map2map grid size (512^3) before computing their summary statistics.

3.3 Summary statistics

We perform our SBI analysis with our muchisimocks framework on three standard summary statistics: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the bispectrum $B_{ggg}(k_1, k_2, k_3)$.

The galaxy–galaxy auto power spectrum $P_{gg}(k)$ is defined by

$$\langle \delta_g(\mathbf{k}) \delta_g(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{gg}(k), \quad (6)$$

where $k = |\mathbf{k}|$ is the Fourier space wavenumber. This is the two-point function of the galaxy field, the most common statistic for cosmological inference from galaxy samples that captures all the Gaussian information. We use 32 logarithmically spaced bins from $k = 0.01$ to $k = 0.4$; the maximum k value is set by the Nyquist frequency of our mocks. We use the BACCO code to compute $P_{gg}(k)$ for our mocks. We can compute the 15 cross-terms of the 5 bias fields and later sum their contributions based on the bias parameters to obtain the power spectrum of the total tracer field; in practice we do this for the noiseless case, but in the noisy case compute $P_{gg}(k)$ for the total tracer field as this would add additional terms.

The galaxy–matter cross power spectrum $P_{gm}(k)$ is defined by

$$\langle \delta_g(\mathbf{k}) \delta_m(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{gm}(k), \quad (7)$$

where $\delta_m(\mathbf{k})$ is the matter density field in Fourier space. This represents the galaxy–galaxy lensing contribution to 3x2pt analyses, and is critical for constraining galaxy bias and breaking parameter degeneracies. We use the same binning as $P_{gg}(k)$, and again use the BACCO code to compute $P_{gm}(k)$ for our mocks; for our fiducial results, we only use up to $k_{\max} = 0.25$ for $P_{gm}(k)$.

The galaxy auto-bispectrum $B_{ggg}(k_1, k_2, k_3)$ is defined by

$$\langle \delta_g(\mathbf{k}_1) \delta_g(\mathbf{k}_2) \delta_g(\mathbf{k}_3) \rangle = (2\pi)^3 \delta_D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B_{ggg}(k_1, k_2, k_3). \quad (8)$$

This is the Fourier space three-point function. We choose 12 linearly spaced bin edges from $k = 0.01$ to $k = 0.4$, yielding 203 triangles; for our fiducial results, we only use triangles with edges up to $k_{\max} = 0.25$, yielding 50 triangles. We use the PolyBin3D code (?) to compute $B_{ggg}(k_1, k_2, k_3)$ for our mocks.

The summary statistics computed on the muchisimocks training set are shown in Fig. 2.

3.4 Implementation and training

Before training, we reparameterize the cosmology and bias parameters to combinations that are better constrained by the summary statistics, and thus should result in improved training convergence (?). *ksf: ask Bea for citation* We multiply the bias and noise parameters by the relevant power of σ_8 , obtaining $\sigma_8 b_1$, $\sigma_8^2 b_2$, $\sigma_8^2 b_{s2}$, $\sigma_8 b_\Delta$, $\sigma_8 A_{b1}^{\text{noise}}$, $\sigma_8^2 A_{b2}^{\text{noise}}$, $\sigma_8^2 A_{b_{s2}}^{\text{noise}}$, and $\sigma_8 A_{b_\Delta}^{\text{noise}}$. When sampling from the posterior, we can divide out the appropriate power of the recovered σ_8 to obtain estimates of the original bias and noise parameters. We also do not include the cosmological parameters h , Ω_b , and n_s in the SBI training, as we do not expect to constrain these parameters

(and confirmed in tests); these are still varied in the training set and implicitly marginalized over.

We use the publicly available `sbi` ? package for the NPE and MAF implementations (??); the MAF uses the `nflows` Python package (?). Our data vector is comprised of the binned statistics discussed in §3.3, concatenated when using multiple statistics. The statistics combinations we show are: $P_{gg}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$, $P_{gg}(k) + P_{gm}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ (we exclude $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$ alone, as these do not contain much constraining power on their own and in practice are almost always considered in combination with $P_{gg}(k)$). We use a 90%/10% training/validation split of our training set. We use the `ADAM` optimizer, and stop training when the loss over the validation loss does not improve for 20 epochs; we save the network weights at the best validation loss. We perform a sweep over several hyperparameters, using 30 random samples from the following sets: the learning rate $\{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}, 10^{-5}\}$, hidden features $\{32, 64, 128, 256\}$, training batch size $\{32, 64, 128, 256\}$, and number of transforms $\{4, 6, 8, 10\}$. The sweeps are performed separately for each statistic or combination of statistics, and for each choice of noiseless (bias parameters only) and noisy (bias and noise parameters); we always use the maximum size training set of 320,000 data vectors, and for the statistics use $k_{\max}^{P_{gg}} = 0.4$, $k_{\max}^{P_{gm}} = 0.25$, and $k_{\max}^B = 0.25$. The model which achieves the minimum best validation loss is selected; we use this best set of hyperparameters to train variations of the model, such as when varying the number of training points and the $k_{\max}$ values. The best hyperparameters for the fiducial $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ noisy case are: learning rate 10^{-4} , hidden feature vector of size 256, batch size 256, and 6 transforms. The set of best hyperparameters vary across the statistics, with the noisy case (which varies more parameters) generally preferring larger hidden feature vectors and batch sizes, and most learning rates around 10^{-4} .

Training on the maximum size training set of 320,000 data vectors takes a few hours on a single CPU for the fiducial model described above. To display the posteriors and compute covariances and other statistics from them, we draw 10,000 samples from the posterior; we confirm that the posteriors are generally converged with this number of samples.

3.5 Test sets

We generate several test sets to validate our muchisimocks suite and our SBI approach.

Our first test set is a *cosmic variance set*. For this we use muchisimocks pipeline to construct 1000 additional mocks each with a different random seeds, all at the same fixed cosmology; we choose this to be the same cosmology as the out-of-distribution SHAME mock test described below. We fix the bias parameters to those measured from that mock, and use these to combine the bias fields into a tracer field for each realization. For the noise parameters, we cannot measure these from the mock directly; to identify realistic noise parameters, we generate 1000 sets of noise parameters and add the resulting noise field to each of the 1000 mocks, obtaining tracer fields that have the same cosmology and bias parameters but different noise parameters. We compute the statistics of these fields, and then compute the reduced χ^2 between them and the statistics of the SHAME mock. Specifically, we use $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ with the fiducial $k_{\max}$ choices (the noise model does not affect $P_{gm}(k)$ as discussed in §2.3 so we exclude it), and choose the set of noise parameters that gives the lowest χ^2 . We then use this set of noise

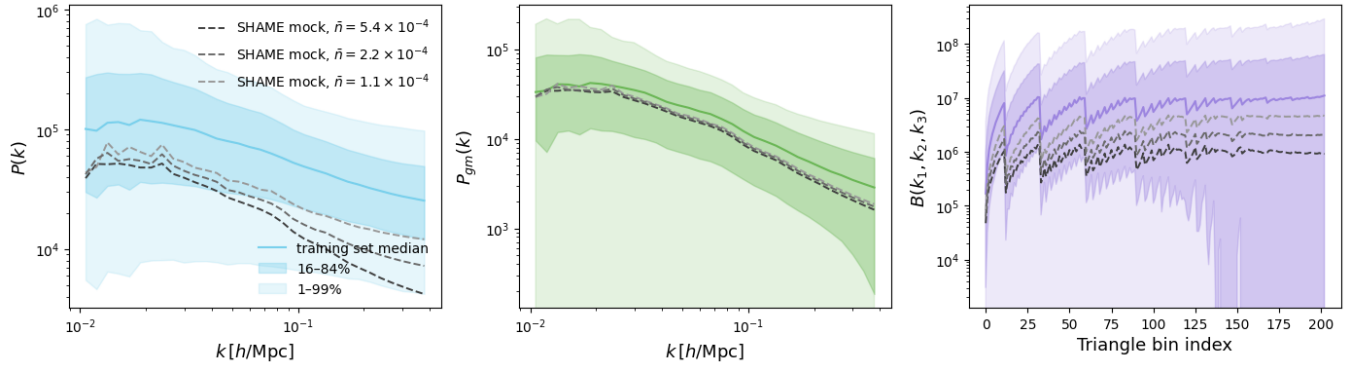


Figure 2. The summary statistics computed on the *muchisimocks* training set: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. The median, 16th–84th percentiles, and 5th–95th percentiles of the training set. The out-of-distribution test mock catalog constructed with extended subhalo abundance matching (SHAME) on a BACCO simulation is also shown (black dashed). In our fiducial inference model, scales below $k_{\max} = 0.25$ for $P_{gm}(k)$ and triangles with sides smaller than $k_{\max} = 0.25$ for $B_{ggg}(k_1, k_2, k_3)$ are excluded. *ksf: legend labels bigger ksf: double check shame; maybe if have b2 that's very large? pgm: avg b1*p11 matches shame. for pk, avg b1**2 * p11 doesn't match. or, addl operators that appear in pgg and not pgm; aka p22. usually it's smaller, but for v broad prior (aka b2 goes up to 2), then part of param space where p22. so could be because of prior. test: do same plot but restrict b2 from -1 to 0.*

parameters and regenerate the noise fields with different noise realizations, and combine this with the biased tracer fields to obtain our final set of 1000 cosmic variance mocks at fixed cosmology, bias, and noise parameters, only varying the random seeds. For the recovery tests, we compute the mean statistic over these 1000 mocks, and perform SBI on that mean to reduce the randomness from cosmic variance; in Appendix A we show that this gives the same result as combining the results of inference on each of the 1000 mocks, while being much faster.

Our second test set is our *coverage set*. This consists of 1000 mocks generated with the *muchisimocks* pipeline that span the same prior range of cosmology, bias, and noise parameters as in the training set. To select the parameters, we construct an independent 1000-point LH for the cosmologies, and a separate LH for the bias and noise parameters. The coverage set tests the performance of our method across the prior space and its dependence on the parameter values.

Our final test set is our *SHAME mock*; we construct a realistic catalog of galaxies using the extended subhalo abundance matching (SHAME) approach applied to a BACCO simulation. This is a stringent test as it uses both a different, significantly higher resolution underlying cosmological simulation and a completely different galaxy formation model than the *muchisimocks* data used to train the inference network, testing the generalizability of our model. In detail, we use the BACCO simulation *ksf: which one?* with a box size of $(1024 h^{-1} \text{Mpc})^3$ (slightly larger than the *muchisimocks* training data) and 3072^3 particles, which uses the fixed-and-paired technique (?), and has cosmological parameters $\Omega_c = 0.309$, $\sigma_8 = 0.816$, $h = 0.677$, $\Omega_b = 0.0486$, and $n_s = 0.967$. Halos are identified using the ? halo finder, and we populate these halos with galaxies using the SHAME model with the following parameter choices: XXX. We choose a fiducial number density of $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$, similar to that of DESI emission line galaxies (ELGs) and slightly lower than DESI luminous red galaxies (LRGs). To check our inference results, we measure the bias parameters of this sample using the field-level probabilistic bias approach of ?; we find $b_1 = 0.474$, $b_2 = 0.127$, and $b_{s,2} = -0.339$ (we do not expect b_Δ to be well-measured by this approach). We also test a higher and lower number density sample, of $5.4 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$ and $1.1 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$; these are measured to have slightly different

bias parameters. The summary statistics computed on the SHAME mock are shown in Fig. 2; they largely fall within the middle 68% of the training set, with the exception of $P_{gg}(k)$ at smaller scales, which reaches the lower 5th percentile but still falls within the training set distribution.

For each of the two simulations of opposite phases from the fixed-and-paired pair, the SHAME model gives a set of discrete galaxy tracers. We deposit these to a mesh using the Cloud-in-Cell mass assignment scheme, of size $(512)^3$. We then remove the high- k modes and deconvolve the field in the same way as the *muchisimocks* fields. We compute the statistics on each of the two fields with the same method used for the training data, and take the mean of these two statistics as the data vector for inference.

4 RESULTS

4.1 In-distribution recovery tests

4.1.1 Cosmic variance test set results

We first validate our SBI pipeline on the cosmic variance test set, at fixed values of the cosmology, bias, and noise parameters. We compute summary statistics on the test data in the same way as on the training data.

For the cosmic variance test set, we compute the mean of all of the statistics computed on the 1000 individual realizations and use this as the data vector for inference. In Appendix A, we show that this recovers the typical size of posterior of the individual samples (and errors wider than the distribution of the means of the individual realizations). This allows us to visualize a typical contour, and average out the effects of cosmic variance to visualize the mean constraint. The posteriors are shown in Figure 3 for three key parameters, Ω_m , σ_8 , and b_1 ; we infer and marginalize over the other bias parameters and the noise parameters. (The full contour plot is shown in Fig. B1.) We find that we obtain tight, unbiased constraints on the cosmology and bias parameters when using the full combination of statistics ($P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$), validating the inference approach.

While $P_{gg}(k)$ on its own provides significant constraining power on Ω_m , it results in a weak constraint on b_1 and a weak, biased

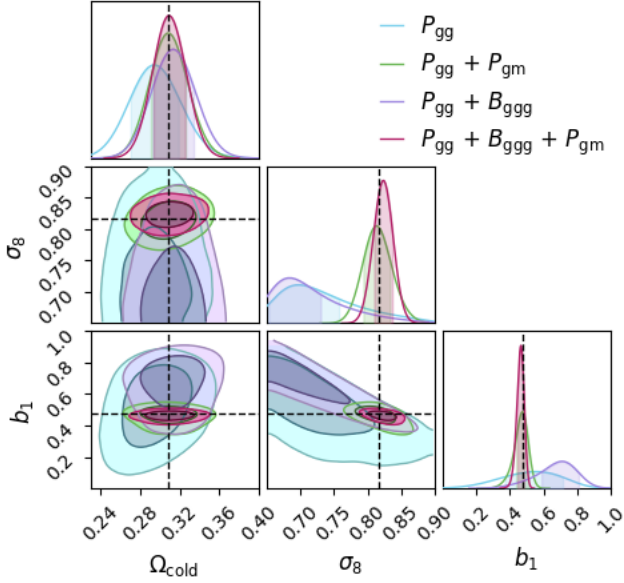


Figure 3. Posteriors for the key parameters Ω_m , σ_8 , and b_1 inferred from the mean of the statistics over our cosmic variance test set, for each of the statistics combinations considered. The inclusion of $P_{gm}(k)$ breaks the $\sigma_8 - b_1$ degeneracy, providing tight and unbiased constraints on the parameters; adding $B_{ggg}(k_1, k_2, k_3)$ tightens the constraints. The other bias parameters and the noise parameters are inferred and marginalized over; the full posterior plot is shown in Fig. B1.

constraint on σ_8 . The weakness of these constraints is expected as a result of the perfect degeneracy between σ_8 and b_1 in the power spectrum (as $P_{gg}(k) = (b_1 \sigma_8)^2 \hat{P}(k)$); we check that the combination $\sigma_8 b_1$ is accurately constrained. Including $B_{ggg}(k_1, k_2, k_3)$ interestingly only slightly increases the precision on Ω_m ; further, it does not improve the bias on σ_8 . Notably, it does significantly increase the precision on the higher-order bias parameters b_2 and b_{s^2} . We see that using the statistics combination $P_{gg}(k) + P_{gm}(k)$ produces unbiased constraints on σ_8 and b_1 ; this makes sense as $P_{gm}(k)$ is able to break this degeneracy (as $P_{gm}(k) = b_1 \sigma_8^2 \hat{P}(k)$). This corroborates our hypothesis that the bias observed in σ_8 for $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ is likely a result of projection effects caused by a lack of constraining power. The full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ preserves the unbiased constraints, and the bispectrum adds an additional improvement of constraining power compared to $P_{gg}(k) + P_{gm}(k)$: it improves Ω_m by XX, σ_8 by XX, and b_1 by XX, and notably the bias parameters b_2 by XX, b_{s^2} by XX, and b_Δ by XX. [ksf: TODO figure out which data to use to quantify this](#)

We note that these results are for real space, and we would expect differences in the relative information content in the statistics in redshift-space, where there is additional cosmological information contained in redshift-space distortions.

4.1.2 Coverage test set results

We infer the parameters of the 1000 samples of our coverage test set, spanning the prior range of our training set; this allows us to test the dependence of the estimated parameters on the true parameter values. Note that the prior has a wide range of values of the noise parameters, so these test samples will have a variety of noise levels.

In Figure 4, we show the differences between the predicted values and true values of the parameters as a function of the true value, which spans the full training prior; we display the median of the 1000 mocks, and the 16-84th percentile range. The predicted value is the mean of the inferred posterior; we check that the behavior is very similar for the maximum a posteriori (MAP) values. For $P_{gg}(k)$, we can see that the range of residuals is relatively wide, especially for b_1 , and for all key parameters show a stronger bias with larger distance from the center of the prior. This is expected when the statistic does not provide significant constraining power on the parameter, as is the case with $P_{gg}(k)$, as the prediction will tend towards the mean. Including $B_{ggg}(k_1, k_2, k_3)$ improves the strength of the parameter-dependent bias for all parameters, and also generally tightens the range of the residuals. When we consider $P_{gg}(k) + P_{gm}(k)$, we can see that the bias on σ_8 improves significantly, as $P_{gm}(k)$ breaks the degeneracy with b_1 as we saw in the cosmic variance test set. $P_{gm}(k)$ also improves the constraints on Ω_m and b_1 . Further including $B_{ggg}(k_1, k_2, k_3)$ helps marginally, and this full parameter combination yields largely unbiased results across the coverage set, indicating that our inference is robust across the prior. There are still slight biases for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ particularly at high Ω_m and σ_8 , so the analysis is most reliable towards the center of the prior.

The residuals shown above only quantify the accuracy of the means of the posteriors; to assess the inferred uncertainties, we compute the coverage of the samples, defined as the fraction of samples for which the true value falls within the predicted uncertainty range, as a function of the given level of uncertainty. For instance, for a credibility level of 68%, 68% of the samples should have the truth fall within their predicted 68% error region (“empirical coverage”). A higher empirical coverage compared to credibility level indicates underconfidence, while lower indicates overconfidence. This coverage (or probability-probability) plot is shown in Figure 5, for the 500 mocks of the coverage set closer to the center of the prior in Ω_m , σ_8 , and Ω_m . We see that the inferred uncertainties are generally quite well calibrated for all three key parameters. Using only $P_{gg}(k)$ or $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a slight underconfidence in the posteriors, at the level of $0.1 - 0.25\sigma$ (assuming a Gaussian distribution). The full statistics combination $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ results in improved calibration of the uncertainties, with only slight fluctuations. We note that for this check, we have excluded the mocks on the edges of the prior to isolate the assessment of the SBI-inferred uncertainties; when we make the coverage plot for all 1000 mocks of the coverage set, we do see some overconfidence. This is a result of the biases towards the edges of the prior which cause the uncertainty regions to not sufficiently cover the truth, and the uncertainty estimates do not fully compensate for this bias.

These results show that our inference pipeline can accurately recover the cosmological and bias parameters across a range of parameter values.

4.2 Out-of-distribution recovery test: SHAMe mock

The results of our SBI pipeline on our SHAMe out-of-distribution test mock with fiducial number density $\bar{n} = XXX$ are shown in Figure 6 for the key parameters (the full contours are shown in Figure B2). We obtain generally unbiased constraints on the cosmological and bias parameters when using the full statistics combination, with Ω_m , σ_8 , and b_1 all within 1σ from the true values. The trends for the statistics generally follow those seen in the cosmic variance test set (§4.1 and Figure 3), with $P_{gm}(k)$ breaking the degeneracy between σ_8 and

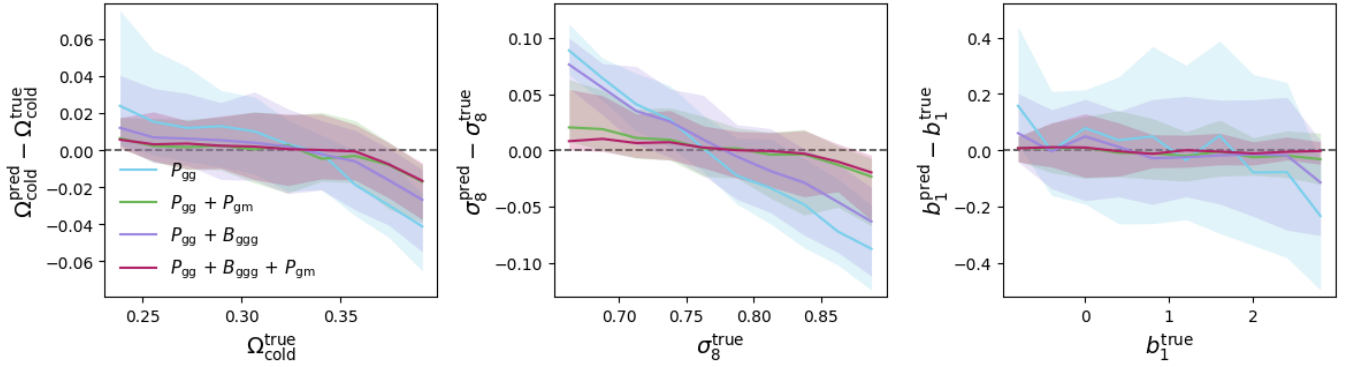


Figure 4. The difference between the predicted (mean of the posterior) and true parameter values for the key parameters for each of the 1000 coverage test set mocks, as a function of the true value. The median and 16th-84th percentile ranges in bins of the true value are shown for each of the statistics combinations considered. Using the full statistics combination gives largely unbiased recovery across the prior range, with a small biases at the high ends of Ω_m and σ_8 ; using fewer statistics results in stronger parameter-dependent biases that tend towards the mean. Including more statistics also results in tighter parameter constraints, most notably with $P_{gm}(k)$ significantly increasing the precision on b_1 across parameter space. *ksf: bias at edges reasonable, prior volume effects dont guarantee this*

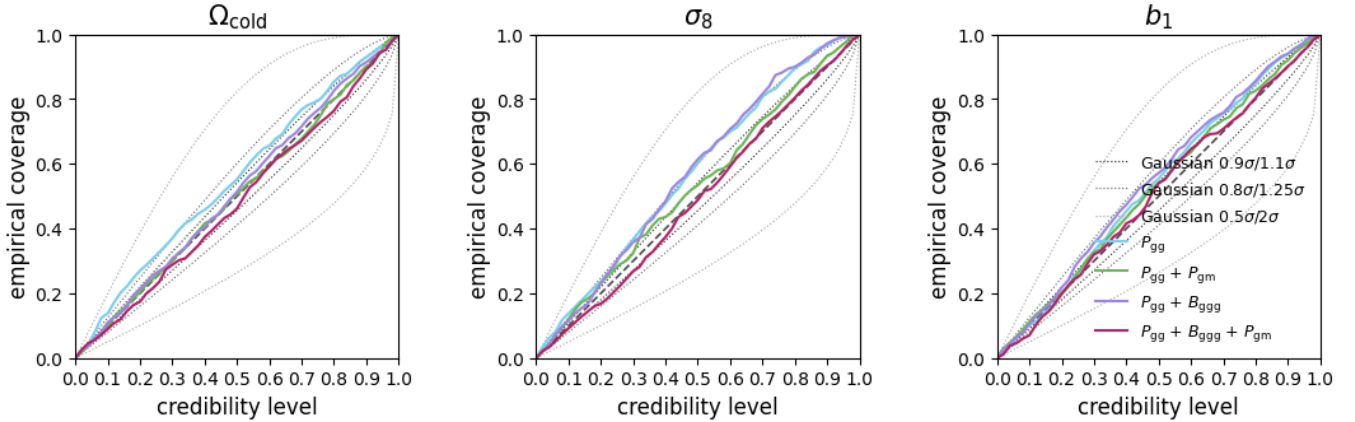


Figure 5. Coverage (probability–probability) plots for the key parameters for the 500 mocks of the coverage test set closest to the centre of the prior. The empirical coverage is the fraction of test set points with predicted uncertainties at the given credibility level that encompass the true value. The closeness to the 1:1 line indicates well-calibrated uncertainties. *ksf: overconfident bc of same issue of fig 5? try with central subspace of cosmology ksf: full combo is worst; doesnt get better adding more stats. ksf: see if gets worse with fewer training sets ksf: see if other sbi papers find similar ksf: marcos isnt too worried, but maybe dep on training data, could try with fewer to see if get worse? ksf: try on combined params ksf: inner subset? see what over papers do, if use coverage set*

b_1 , and $B_{ggg}(k_1, k_2, k_3)$ improving constraints significantly for the bias parameters and moderately for the cosmological parameters. For the SHAME data, we obtain a XXX% improvement in precision on Ω_m and XXX% on σ_8 when including $P_{gm}(k)$ compared to $P_{gg}(k)$ alone; the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ results in XXX% and XXX% improvements on Ω_m and σ_8 respectively compared to $P_{gg}(k)$.

We do see slight biases in the values of b_1 and b_2 when using the full statistics combination, at the XXX σ and XXX σ levels, while these were accurately constrained on average in the cosmic variance test set. This is likely a result of the bias parameters absorbing some model mismatch, which could be an issue for approaches such as imposing tighter priors on bias parameters; however, for our inference the cosmological parameters remain unbiased.

We show the results for the full parameter combination on the SHAME samples with a range of number densities in Figure 7. We find that the constraints on the cosmological parameters are robust to

the noise level across this range. The constraints on the bias parameter b_1 do lose precision at lower number densities; this could be related to degeneracies with the noise parameters. We note that the noise parameters themselves are poorly constrained, as seen in the full contour plot (Figure B2). However, we check that the noise model is necessary for properly inferring the parameters; assuming noiseless training data results in strong biases in the inference.

This test on out-of-distribution data demonstrates the robustness of our forward model to modeling realistic galaxy populations constructed with completely different models.

4.3 Scale dependence

We investigate the dependence of our results on the maximum k scale included in our inference, for each of the statistics. We keep the maximum scale for $P_{gg}(k)$ fixed to $k_{\max} = 0.4 \, h\text{Mpc}^{-1}$, the Nyquist frequency of the box. At this scale, the map2map emulator of

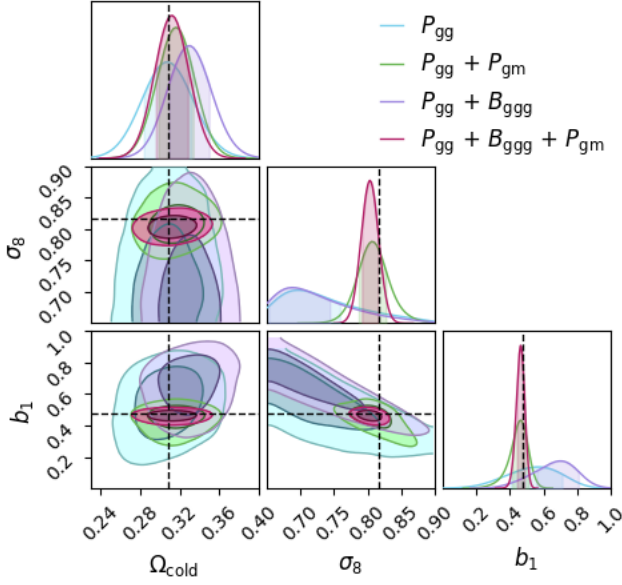


Figure 6. Posteriors for the key parameters Ω_m , σ_8 , and b_1 for our out-of-distribution SHAMe mock test, for each of the statistics combinations considered. The recovered parameters with the full statistics combination are unbiased within 1σ , and the statistics show similar constraining power as in the in-distribution test. The full posterior plot is shown in Fig. B2. ksf: bispec does add a lot to sigma8; 30% even. important enough to say in abstract

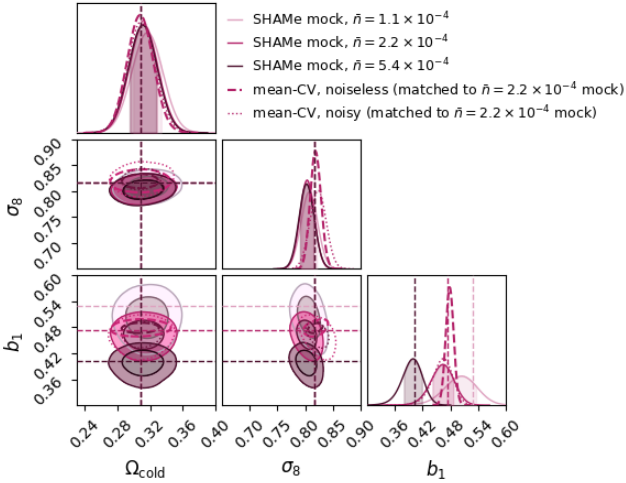


Figure 7. Impact of noise on the inferred posteriors, for the mean of $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ over the cosmic variance test set. The brown contours show inference trained on statistics computed on noiseless realizations, and inferred with a model with no noise; the pink contours show inference computed on realizations injected with the rubiraschmidnoise2025 ksf: update model, and inferred with the same noise model. Even with perfect knowledge of the noise model, the constraints on σ_8 and b_1 degrade significantly. ksf: should i show what happens if you try to infer on noisy data with a noiseless model? (it gets real biased) ksf: could be interesting to show what happens w denser model. complement shame at diff number densities - add that plot! did lrg-like same, but may make sense to have a much denser sample - more to gain. ksf: esp bc limited by noise model for pgm

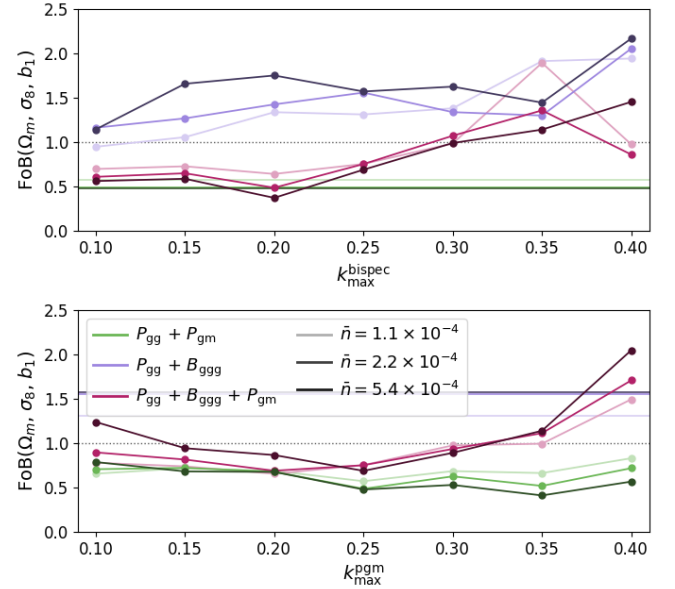


Figure 8. Scale dependence of the figure of bias (FoB) for the three key parameters combined for the SHAMe out-of-distribution mock. The top panel shows the FoB as a function of $k_{\max}$ for the bispectrum, fixing to our fiducial $k_{\max} = 0.25$ for $P_{gm}(k)$; the bottom panel shows it as a function of $k_{\max}$ for $P_{gm}(k)$, fixing to our fiducial $k_{\max} = 0.25$ for the bispectrum. In both cases, the FoB for the full statistics combination remains around or below 1σ until $k_{\max} = 0.25$, after which the inference becomes biased. ksf: fluke that goes back down at 0.4; so close to nyquist freq. could try 0.37 for pgm ksf: try restricting pk to 0.35, @fid values of others. ksf: remove 0.4 from this plot? ksf: slight bias for pgg+b coming from projection effects? could make this plot for cosmic variance set... maybe just mention that these are vol effects. or look at the params individually. incl b1 times sig8. could add a param orthogonal to that product; sigma8/b1. ksf: lowest 2 pink points on top plot confusing that they go above green points. look at contour plots ksf: maybe because noise model not affecting pgm! ksf: marcos: at low kmax end, looks like noise when still dominated by priors. noise parameters are close to edges? ksf: consistent with marcos's paper - with just pgg+bgg, no clear break. only when include matter. ksf:

N -body displacements is shown to achieve significantly sub-percent accuracy ($\sim 0.2\%$) on the power spectrum (?), and ? showed that map2map combined with the hybrid Lagrangian bias expansion to second order achieves 1-2% precision to $k_{\max} = 0.6 h\text{Mpc}^{-1}$. (Previous works using the hybrid bias expansion with a power spectrum emulator find accuracy down to $k_{\max} = 0.6 h\text{Mpc}^{-1}$ (??).) For the bispectrum, ? recently validated the hybrid bias expansion down to $k_{\max} = 0.25 h\text{Mpc}^{-1}$. ksf: anything to cite/mention for the cross-power-spectrum?

In Figure 8, we show the figure of bias for the SHAMe out-of-distribution mock for the three key parameters Ω_m , σ_8 , and b_1 combined, as a function of the $k_{\max}$ of $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$; when varying $k_{\max}^{B_{ggg}}$, we fix $k_{\max}^{P_{gm}} = 0.25 h\text{Mpc}^{-1}$, and when varying $k_{\max}^{P_{gm}}$, we fix $k_{\max}^{B_{ggg}} = 0.25 h\text{Mpc}^{-1}$. We see that using only $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a bias moderately above 1σ for all number density samples, regardless of $k_{\max}$; this is likely largely a result of projection effects. When including the cross spectrum, $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, the inference is accurate to 1σ up until $k_{\max}^{B_{ggg}} = 0.25 h\text{Mpc}^{-1}$, aligning with the result validating the hybrid bias expansion on the bispectrum and indicating that the limiting factor is the expansion rather than the emulated displace-

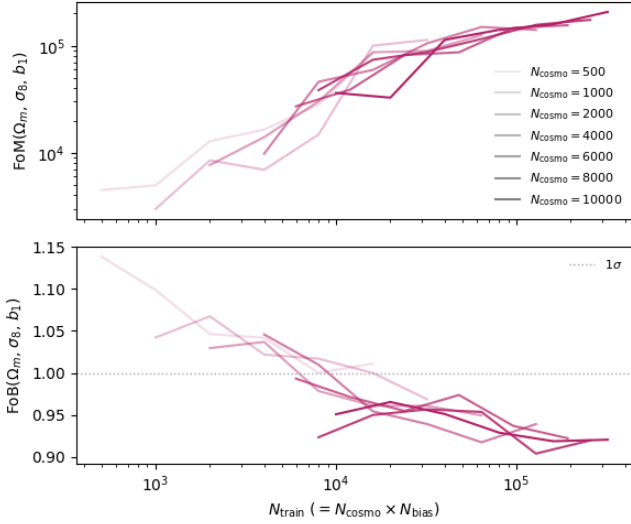


Figure 9. Convergence of the inference with training set size, for $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ on the coverage test set for the three key parameters Ω_m , σ_8 , and b_1 combined. The top panel shows the figure of merit (FoM); the bottom panel shows the figure of bias (FoB), normalized by the 68th percentile of the χ^2 distribution for three dimensions (corresponding to the three parameters). The opacity indicates the number of training cosmologies. Increasing the number of training points increases the precision significantly until $N_{\text{train}} \sim 20,000$ and then more marginally, with the number of cosmologies and bias parameter sets contributing similar precision; the bias also plateaus around $N_{\text{train}} \sim 20,000$, with the number of cosmologies mattering more than the number of bias parameters for smaller N_{train} . [ksf: higher res](#) [ksf: say median/mean of coverage set](#)

ments. The statistics combination $P_{gg}(k) + P_{gm}(k)$ results in accurate inference for all number densities through $k_{\text{max}}^{\text{Pgm}} = 0.4 \, h\text{Mpc}^{-1}$. However, when additionally adding $B_{ggg}(k_1, k_2, k_3)$, the inference becomes biased pushing below $k_{\text{max}}^{\text{Pgm}} = 0.25 \, h\text{Mpc}^{-1}$. We also check the full grid of combinations, and find $k_{\text{max}}^{\text{Bggg}} = 0.25 \, h\text{Mpc}^{-1}$, $k_{\text{max}}^{\text{Pgm}} = 0.25 \, h\text{Mpc}^{-1}$ to be the optimal combination of limiting scales; pushing either or both to smaller scales results in further bias. It is initially odd that pushing $P_{gm}(k)$ to smaller scales results in a bias only when including the $B_{ggg}(k_1, k_2, k_3)$; we speculate that this could be because our noise model does not account for noise in $P_{gm}(k)$, though further investigation is merited.

4.4 Dependence on number of cosmologies and bias parameter sets

We test the convergence of our inference with the number of training data points. For varying the number of cosmologies, we train the SBI network on a subset of our cosmologies. To vary the number of bias parameters, we populate each of those cosmologies with a varying number of bias parameter sets. We compute the mean squared error (MSE) over all the parameters for the coverage set samples; these errors as a function of the number of training points is shown in Figure 9.

We see that ...

5 DISCUSSION

We also tested other SBI approaches, including Moment Networks. However, we found these to be...

There are many possible extensions to the model. We could extend the bias model to the third order, which has been shown to be important when... We could even generate larger volumes using map2map; volume is a main limitation of current SBI approaches.

There are many further potential uses of the *muchisimocks* suite. For instance, one could learn to reconstruct the velocity fields given the density fields. The suite could be used to explore the information in higher-order statistics of the density field, including optimal neural network compressions. One could also investigate the effect of more restrictive priors on the Lagrangian bias parameters, encoding our knowledge specific galaxy samples.

The smallest scales we can include are limited by... [ksf: when limited by map2map vs HBE?](#)

6 SUMMARY AND CONCLUSIONS

ACKNOWLEDGEMENTS

The authors would like to thank Beatriz Tucci, Matt Ho, and Jed Homer; Matteo Zennaro, Daniel López-Cano, Francisco Maion and the rest of the BACCO group; and Risa Wechsler and the KIPAC Galaxy Formation and Cosmology group for helpful discussions.

DATA AVAILABILITY

The *muchisimocks* suite and the code used in this paper will be made publicly available at [thisurl](#).

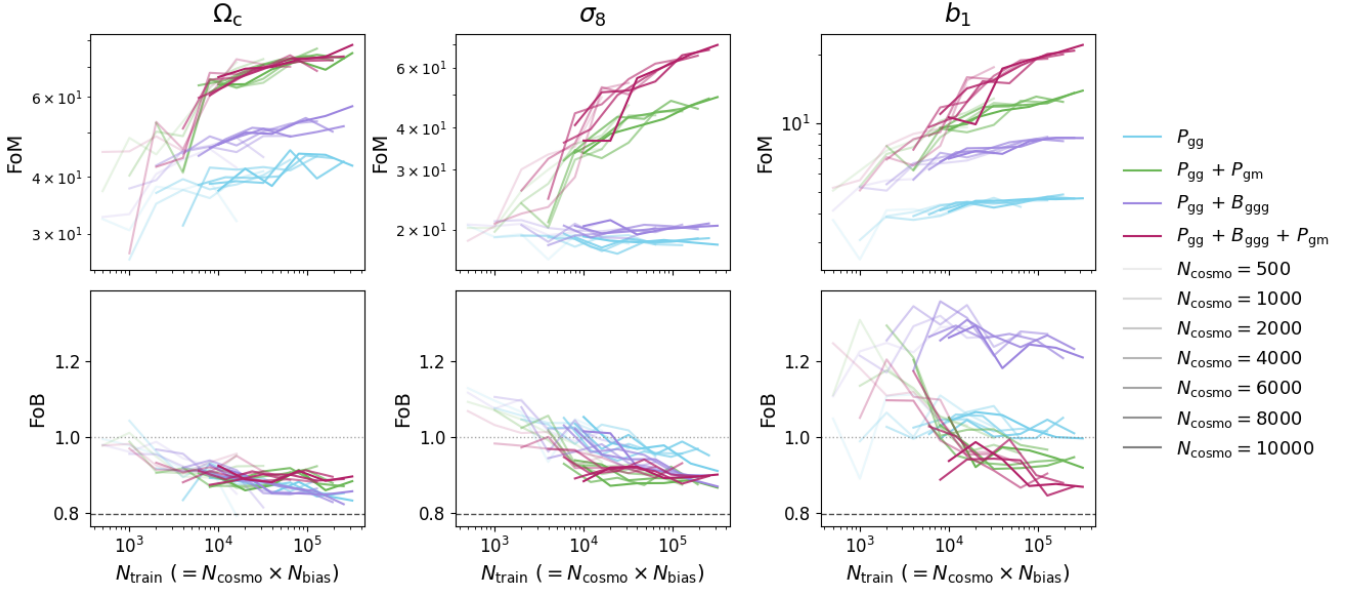


Figure 10. Convergence of the inference with training set size, for all the statistics combinations on the coverage test set for the three key parameters Ω_m , σ_8 , and b_1 individually. The top row shows the figure of merit (FoM) and the bottom row shows the figure of bias (FoB); The opacity indicates the number of training cosmologies, and the dashed line indicates the expectation for an unbiased Gaussian distribution. The FoM increases with training set size for most statistics and parameters, with the exception of σ_8 with $P_{gg}(k)$ and $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ which do not show improvement; all have an FoB around or below 1σ . ksf: not reaching 0.8 because network not good enough? non-gaussian? ksf: try with cosmic variance set? ksf: show b1, not mult with b1 ksf: reach theory floor bc have addl source of noise; network + data vector, projection effects ksf: om probs comes mostly for geometric aspects of power spec (??) ksf: takeaway is that many are still not converged. ref how many sims paper - but diff research question, number for emulator vs sbi?

REFERENCES

APPENDIX A: IMPACT OF VARIANCE AND NUMBER DENSITY

We investigate the behavior of the posteriors of the cosmic variance test set, which contains 1000 realizations run with different seeds at the same cosmology, bias, and noise parameters. Our goal is to be able to quantify and visualize the accuracy and precision of the inference on this test set.

We first compute the statistics on all 1000 realizations, and perform the inference on each individually with our SBI pipeline. We obtain a posterior for each of the 1000 realizations. We then look at the distribution of the means of these posteriors. To visualize the error regions, we compute the mean-of-the-means of the posteriors, and shift each posterior such that it is centered on that value; we then pool the posteriors, drawing 500 samples from each. This shows the typical size and shape of the error region, centered on the mean.

We also compute the mean of the statistics of the 1000 mocks, and perform the inference on that mean data vector. This mean data vector has the benefit of reducing the effect of cosmic variance on the statistic, but brings the complication of having a different covariance than the data the SBI model was trained on (namely, much lower variance). In standard MCMC approaches, this would be handled by appropriately scaling the covariance matrix, but in the context of SBI, the covariance is learned by the model from the training data and is not given explicitly. Thus it is important to understand the behavior of the model when we use the mean data vector for inference.

Figure A1 shows the result of these three quantities on the cosmic variance test set, all for the combined statistic $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$. The black contour shows the distribution of the means of the 1000 posteriors; the grey contour shows the recentered, pooled posteriors; and the magenta contour shows the inference per-

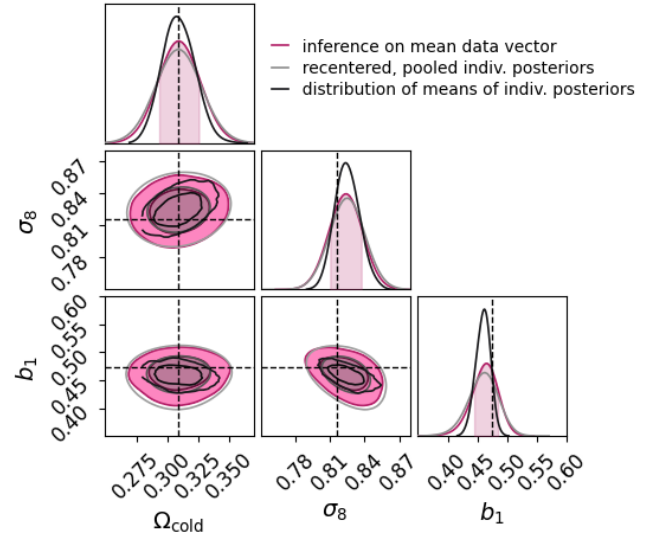


Figure A1. Validation of inference on the mean of the statistics for the cosmic variance test set, for the full $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ combination. The grey contours show 10 posteriors of inference on statistics randomly selected realizations from the test set, and the pink contours show inference on the mean of 1000 statistics over all the realizations. The shape and width of the posteriors are consistent, with the mean falling in the middle of the individual posteriors. ksf: bc network is focusing on low dim representation ksf: in each projection, get peak of dist (or mean); and plot 2d hist of the grey means. and the 1ds, will be a hist of grey vs the contours ksf: dont need to focus on uncertainties ksf: importance sampling of the posteriors of individuals - interpolate to get posterior value for each point in param space. then could combine. can share data with marcos ksf: test with fewer than 1000 to see if an issue with number of points

formed on the mean of the 1000 data vectors. We see that the mean-of-the-means inference lies nearly exactly on top of the recentered, pooled inferences. This indicates that despite the lower variance of the mean data vector, the SBI pipeline produces a posterior that has the typical width of a posterior estimated with the noisier data vector, as that was the noise level of data it was trained on. We thus use the inference on the mean data vector throughout the paper to be able to show the typical result of the inference, with the effect of cosmic variance (which would cause a bias in any given individual sample) removed.

We also see that the distribution of the means of the individual realizations is significantly tighter than the width of a typical individual posterior, especially for b_1 . This indicates that our inference is underconfident, predicting generally broader posteriors than would a perfectly calibrated model. This should be taken into account when interpreting the results of this analysis. However, the mean of distribution of means agrees with the true values of the parameters, as well as with the inference on the mean, indicating that our results are unbiased.

ksf: i am not quite sure how to square the results of these two plots. when we perform inference on the mean of 1000, we get the same posterior size as any individual. but when we use the higher number density shame model, we do get a smaller posterior size compared to the lower number density one. (i think the noiseless one makes a bit more sense, as that's actually a different model where we also train on noiseless data. but the different noise levels of shame are all using the same model with the same training data.)

APPENDIX B: FULL POSTERIOR CONTOURS

This paper has been typeset from a $\text{\LaTeX}$ file prepared by the author.

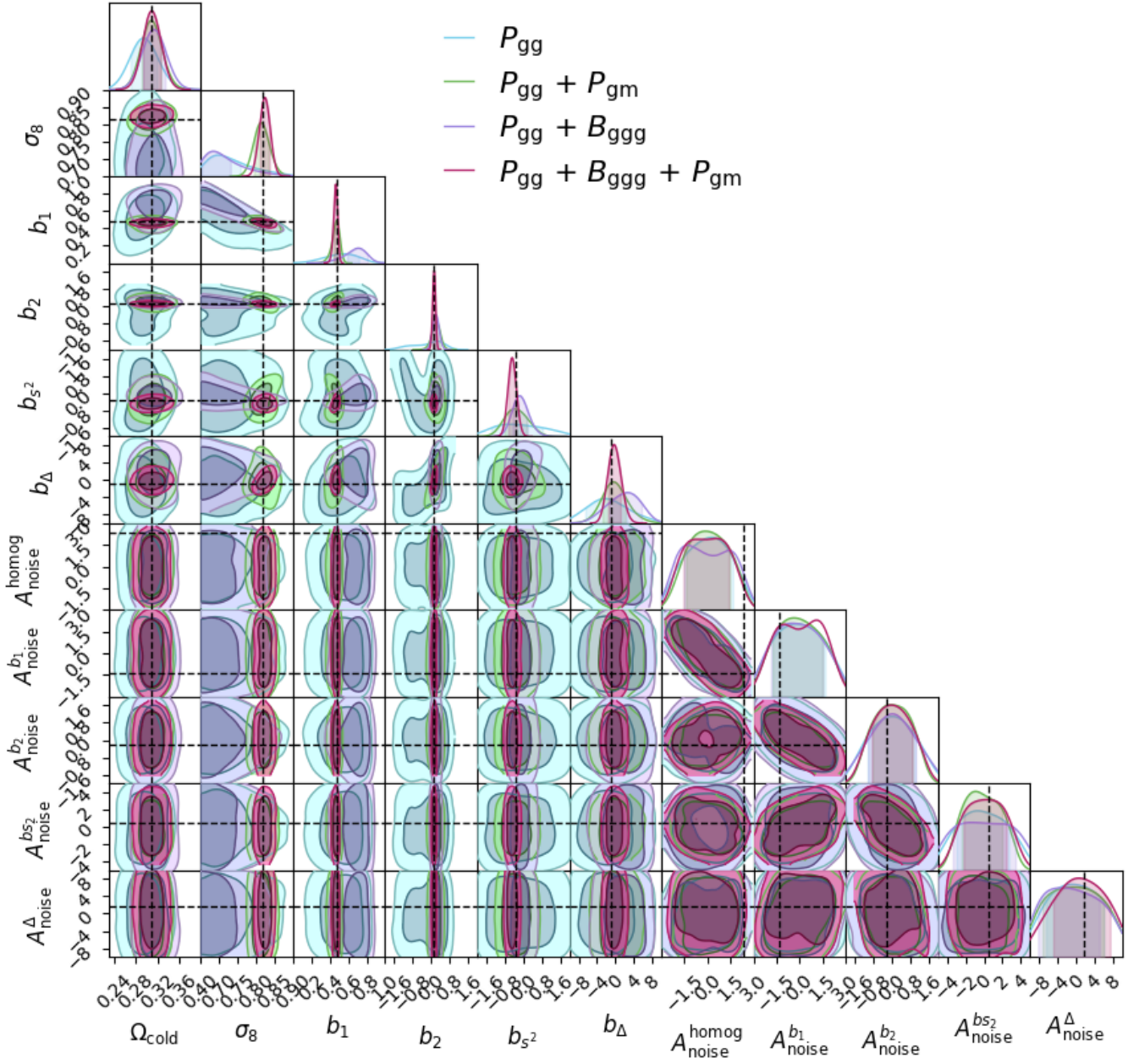


Figure B1. Full posterior contours for the cosmic variance test set.

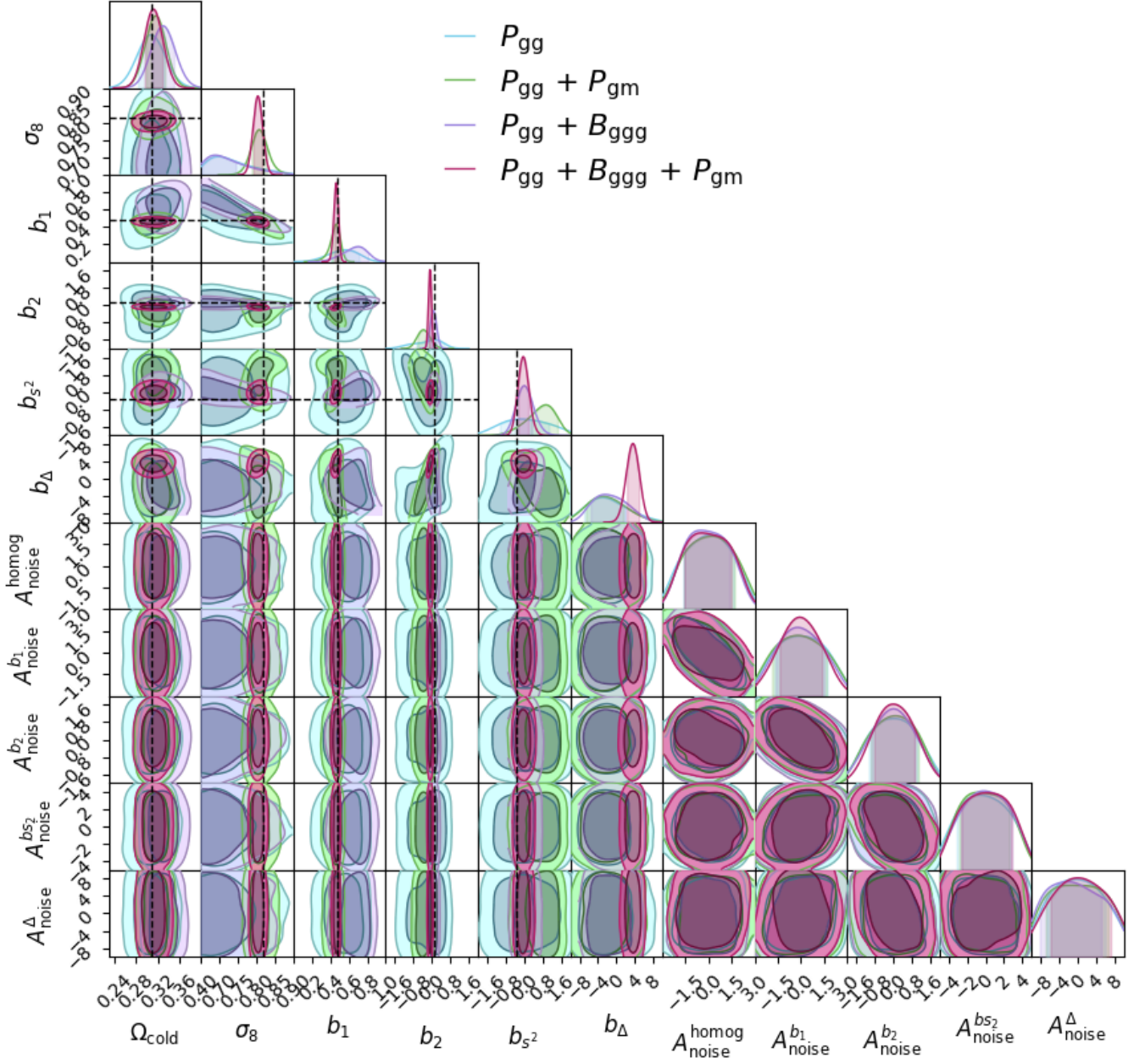


Figure B2. Full posterior contours for the SHAME test set.